

Non-occupational malignant pleural mesothelioma due to asbestos and non-asbestos fibres.

BACKGROUND AND AIM: The occurrence of malignant pleural mesothelioma (MPM) has been reported among population groups with no documented professional exposure to asbestos fibres living in different geographic areas. This paper reviews existing data related to non occupational MPM including its occurrence in the province of Catania (Sicily, Italy). **METHODS:** An electronic search of literature related to non occupational MPM was performed including the year 2005. **RESULTS:** Non occupational MPM in subjects living in areas contaminated by a variety of asbestos and non asbestos fibres has been well documented through a number of epidemiologic studies including cases series, case-control studies, and a cohort study. In addition, the observation of familial clustering of MPM, suggests that genetic factors may play a role in the pathogenesis of this malignancy. The epidemiological evidence also suggests that MPM may occur as a result of the interaction between environmental carcinogens, genetic factors, and virus infection. **CONCLUSION:** It is likely that genetic predisposition and non-occupational exposure to low doses of asbestos and asbestos-like fibres may concur to the development of malignant mesothelioma. However, additional epidemiological and laboratory studies are needed to further understand the relationship between environmental exposure and individual susceptibility to this malignancy.